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Pleistocene Gobiid Fishes of the Genus *Rhinogobius* from Kusu Basin, Oita Prefecture, Japan

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Abstract Two species of Pleistocene gobiid fishes, *Rhinogobius giurinus* and *R. brunneus*, were collected from the diatomite bed at Kusu Basin in Oita Prefecture, Japan. These species are identified by characters of the pelvic fin and scales. From Kusu Basin, a salmonid species, *Oncorhyncus* cf. *O. masou* or *O. rhodurus*, and three cyprinid species *Hemibarbus barbus*, *Zacco temminckii*, and *Acheilognathus lanceolata* were reported by UYENO *et al.* (1975). This is the first record of gobiid fishes which were found from this deposit.

A number of fossil specimens belonging to the family Gobiidae were collected from Nogami Member in Kusu Formation at Nogami in Kusu Basin, Oita Prefecture. This Formation and Member were first described by SHUTO (1953) who made a geological survey of the upper area of Kusu River, and named it as Kusu Formation and recognized three Members: Hōsenji, Nogami, and Hijiu Member in order from bottom to top. The age of Nogami Member is Middle Pleistocene (IWAUCHI and HASE, 1987). UYENO, *et al.* (1975) reported freshwater fish fossils from Nogami Member. These are a salmonid species, *Oncorhyncus* cf. *O. masou* or *O. rhodurus*, and three cyprinid species, *Hemibarbus barbus*, *Zacco temminckii*, and *Acheilognathus lanceolata*. Two species of gobiid fossil fishes of the genus *Rhinogobius* are reported here for the first time from Kusu Basin.

Acknowledgments

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Class Osteichthyes
Order Perciformes
Family Gobiidae
Rhinogobius giurinus (RUTTER)

Material: Specimens were collected by Akinori TAKAYAMA, Teruya UYENO, and Yoshitaka YABUMOTO between 1983–1986.

KMNH (Kitakyushu Museum of Natural History) VP 100,117, almost complete specimen with dorsal side exposed, but first dorsal fin and anal fin missing. Head region well preserved. Standard length 19.8 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the pelvic fin. KMNH VP 100,118, almost complete specimen with its left side exposed, but second dorsal fin missing. Scales of the posterior part of the body are well preserved. Standard length 24.8 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,119, almost complete specimen with ventral side exposed, but first dorsal fin, a part of 2nd dorsal fin and pelvic fin missing. Standard length 20.6 mm and number of vertebrae 10 +16=26. This specimen is identified by the form of the scales. KMNH VP 100,120, almost complete specimen with its right side exposed, but lacking dorsal fin, a part of anal fin and some bones of head region. Standard length 26.1 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the pelvic fin. KMNH VP 100,121, almost complete specimen with its right side exposed, but lacking pectoral fin, pelvic fin, anal fin and a part of head region. Scales are preserved on abdominal part and are detached from caudal part. Standard length 25.6 mm. This specimen is identified by the form of the scales. KMNH VP 100,122, almost complete specimen with its right side exposed. Scales are preserved. Standard length 26.0 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,123, anterior part of the body and caudal fin with its left side exposed, scales preserved on the part below second dorsal fin. Standard length 24.1 mm. This specimen is identified by the form of the scales. Takayama's specimen 1, almost complete specimen with its left side exposed, but lacking a part of head region and anal fin. Scales are preserved. Standard length 27.0 mm and number of vertebrae 10+16= 26. This specimen is identified by the form of the scales. KMNH VP 100,124, some bones of head region, pectoral fin, dorsal spines, and scales are preserved. This specimen is identified by the form of the scales.

Distinguishing characters: The branching point of the innermost pelvic fin ray is approximately in between the base and the branching point of the next ray. The posterior end of each scale is slightly pointed. The spines of each scale are short at the posterior corner and gradually become longer toward the dorsal and ventral ends of the posterior margin. Grooves extend close to the posterior margin of the scales.

Description: Scales are ctenoid. The frontal bone is wide at the posterior part and rapidly narrows at the anterior part. The sensory canal is present on the lateral margin

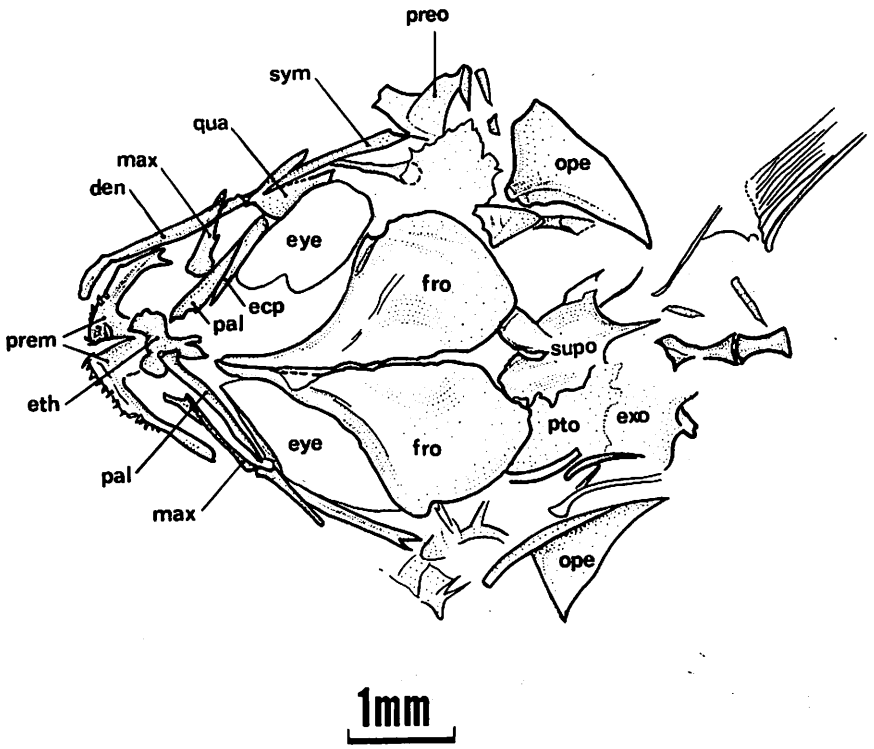

Fig. 1. Dorsal view of head region of a Pleistocene *Rhinogobius giurinus*, KMNH VP 100,117. den, dentary; ecp, ectopterygoid; eth, ethmoid; exo, exoccipital; fro, frontal; max, maxillary; ope, opercle; pal, palatine; prem, premaxillary; preo, preopercle; pto, pterotic; qua, quadrate; supo, supraoccipital; sym, symplectic.

of the frontal and opens at the lateral end. The melanin of the eye ball is preserved in KMNH VP 100,117 (Fig. 1). The dentary is narrow at the anterior part and deep at the posterior part. Small unicuspid teeth are present on the premaxillary and the dentary (Fig. 1). The ceratohyal is narrow at the anterior part and wide at the posterior part. Four branchiostegals are attached on the ceratohyal. The epihyal is almost triangular in shape. The posterior margin of the pectoral fin is circular. Other observable bones of the head region are maxillary, quadrate, symplectic, palatine, ectopterygoid, opercle, preopercle, supraoccipital, pterotic and exoccipital (Fig. 1). The first and second and the third and fourth hypurals are fused. The parhypural is short. The haemal spine of the second preuralcentrum is thick and wide. Other observable bones of the caudal region are a part of the 5th hypural and a part of the epural (Fig. 2).

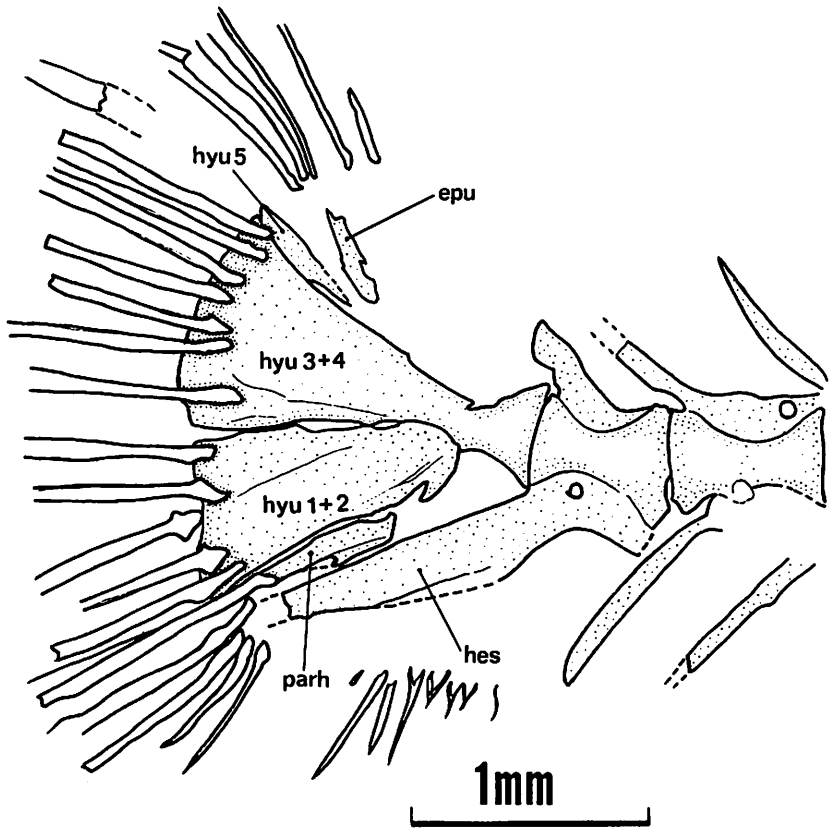

Fig. 2. Caudal region of a Pleistocene *Rhinogobius giurinus*, KMNH VP 100,122. epu, epural; hes, haemal spine of the second preuralcentrum; hyu, hypural; parh, parhypural.

Rhinogobius brunneus (TEMMINCK et SCHLEGEL)

Material: KMNH VP 100,125, almost complete specimen with its left side exposed. Standard length 26.0 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the pelvic fin. KMNH VP 100,126, almost complete specimen with its right side exposed. Standard length 24.3 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,127, body twisted at anal origin, all elements of skeleton, fin rays and scales are detached. This specimen is identified by the form of the scales. KMNH VP 100,128, almost complete specimen with its left side exposed, but ribs, pectoral, pelvic and anal fins missing. Standard length 23.7 mm. This specimen is identified by the form of the scales. KMNH VP 100,129, almost complete specimen with its left side exposed, but pelvic fins missing. Standard length 35.8 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,130, specimen with its left

side exposed lacking first dorsal fin, caudal skeleton and caudal fin. This specimen is identified by the form of the scales. KMNH VP 100,131, specimen with ventral side exposed lacking caudal part. Some bones of head region preserved. This specimen is identified by the form of the scales. Takayama's specimen 2, almost complete specimen with its left side exposed, but first dorsal fin missing. Some bones of head region disarticulated. Standard length 37.5 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,132, almost complete specimen with its right side exposed, but a part of the anal fin missing. Standard length 37.4 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,133, almost complete specimen with its left side exposed, and some bones of head region disarticulated. Standard length 27.0 mm and number of vertebrae 10+16=26. This specimen is identified by the form of the scales. KMNH VP 100,134, almost complete specimen with its left side exposed, but lacking some bones of head region and a part of abdominal vertebrae. This specimen is identified by the form of the scales. KMNH VP 100,135, caudal part of the body. This specimen is identified by the form of the scales. KMNH VP 100,136, anterior part of the body. This specimen is identified by the form of the scales. KMNH VP 100,137, caudal part of the body, lacking upper part of the caudal skeleton. This specimen is identified by the form of the scales.

Distinguishing characters: The branching point of the innermost pelvic fin ray is beside the branching point of the next ray. The posterior margin of each scale is round in shape. The spines of each scale are almost equal in length and grooves extend two

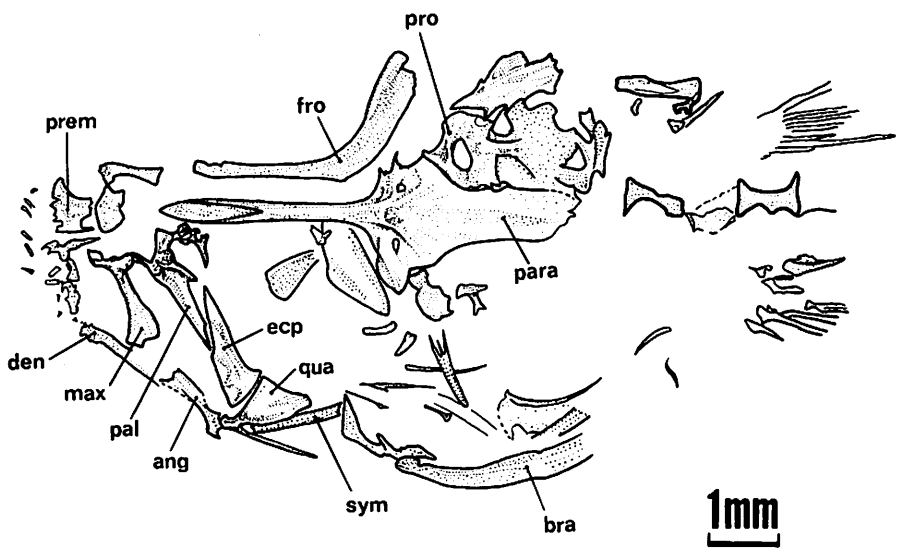

Fig. 3. Ventral view of head region of a Pleistocene *Rhinogobius brunneus*, KMNH VP 100,131. ang, angular; bra, branchiostegal; para, parashenoid; pro, prootic; other abbreviations see Fig. 1.

thirds of the way from the anterior end.

Description: Scales are ctenoid. In the fossil specimens, the observable bones of the head region are maxillary, premaxillary, dentary, angular, palatine, ectopterygoid, quadrate, symplectic, frontal, parasphenoid, prootic, and branchiostegals. There are two condyles on the anterior end of the maxillary. The middle part of the maxillary is broad. Small unicuspid teeth are present on the premaxillary and the dentary. The palatine has two processes and one acetabulum at the anterior end. The posterior margin connects with the ectopterygoid which is pointed at the upper end. The lower end of the ectopterygoid is wide and connects with the anterior margin of the quadrate which is approximately triangular with a long process ventrally. The antero-ventral corner of the quadrate is a condyle for the angular (Fig. 3). A pair of foramina is present at the middle of the parasphenoid. The anterior quarter of the ventral surface of the parasphenoid is slightly concave for the attachment to the prevomer. The prootic has a large foramen for the facial nerve (Fig. 3). Forms of these observable bones are similar to those of Recent *R. brunneus*.

Remarks: In the Recent species of the genus *Rhinogobius*, the first pair of normal long ribs are attached to the third vertebra. The number of dorsal pterygiophores correspond to the number of dorsal fin rays. The space between the fifth and sixth spines of the first dorsal fin is much wider than spaces between other spines of the fin. The number of anal pterygiophores is one fewer than that of anal fin rays. The haemal spine of the first caudal vertebra is conspicuous (Fig. 6). This information is useful in determining the numbers of abdominal and caudal vertebrae, and the numbers of dorsal and anal fin rays in *Rhinogobius* fossil material.

The following characters indicate that the fossil specimens are members of the teleostean fish family Gobiidae of the order Perciformes: (1) the first and second dorsal fins are present and are not continuous; (2) both pelvic fins are close together and situated below the pectoral fin; (3) the body is covered with ctenoid scale; (4) the hypural bones are fused and simplified; (5) the dentary and premaxillary bear small conical teeth; (6) the interorbital space is narrow. They have the following meristic characters: (1) the number of abdominal vertebrae is 10 and the number of caudal vertebrae is 16; (2) the first dorsal fin consists of 6 spines and the second dorsal fin consists of one spine and 8 soft rays; (3) the anal fin also consists of one spine and 8 soft rays. According to MASUDA *et al.* (1984), 35 species in 18 genera of gobiid fishes are found in freshwater in Japan. Among them, the meristic characters of *Rhinogobius giurinus* and *R. brunneus* correspond to that of the fossil specimens. Three species of the genus *Rhinogobius* have been reported in Japan. The fossil specimens are not *Rhinogobius flumineus*, because *R. flumineus* has 11 abdominal vertebrae. The differences between Recent *R. giurinus* and *R. brunneus* are recognized in the forms of the sucking disk of the pelvic fin and scales. UYENO and IWAO (1966) described the differences in the shape of the sucking disk and the number of branches of the innermost pelvic fin ray. But these characters of the pelvic fin are not

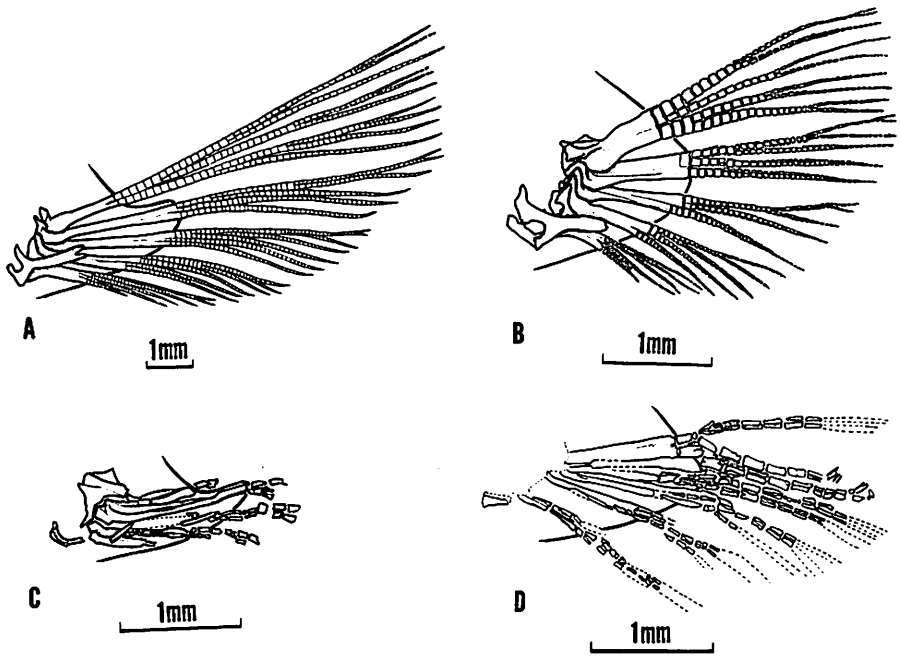

Fig. 4. Pelvic fins of Recent and fossil specimens of the genus *Rhinogobius*. A, Recent *R. giurinus*; B, Recent *R. brunneus*; C, fossil *R. giurinus*, KMNH VP 100,118; D, fossil *R. brunneus*, KMNH VP 100,125. The lines indicate the branching points of fin rays.

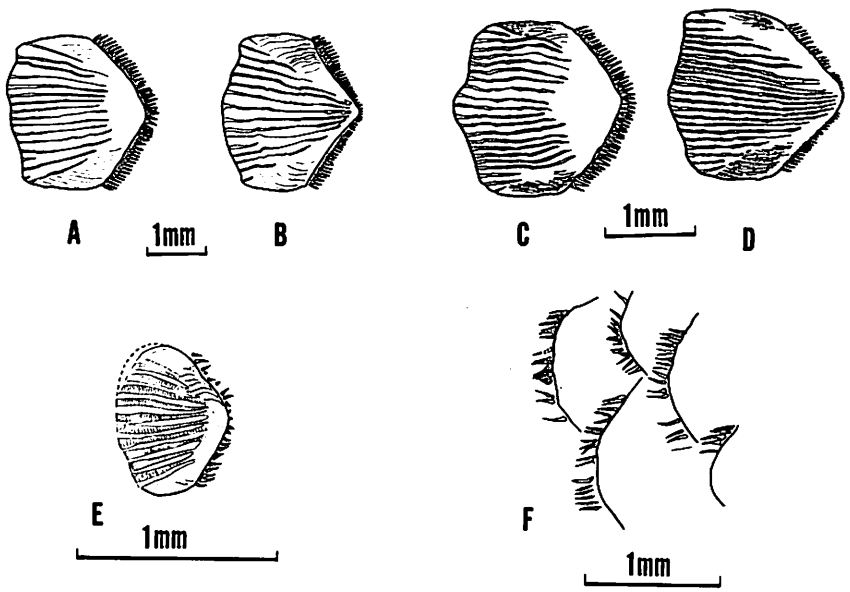

Fig. 5. Scales of Recent and fossil specimens of the genus *Rhinogobius*. A and B, Recent *R. giurinus*; C and D, Recent *R. brunneus*; E, fossil *R. giurinus*, KMNH VP 100,118; F, fossil *R. brunneus*, KMNH VP 100,132.

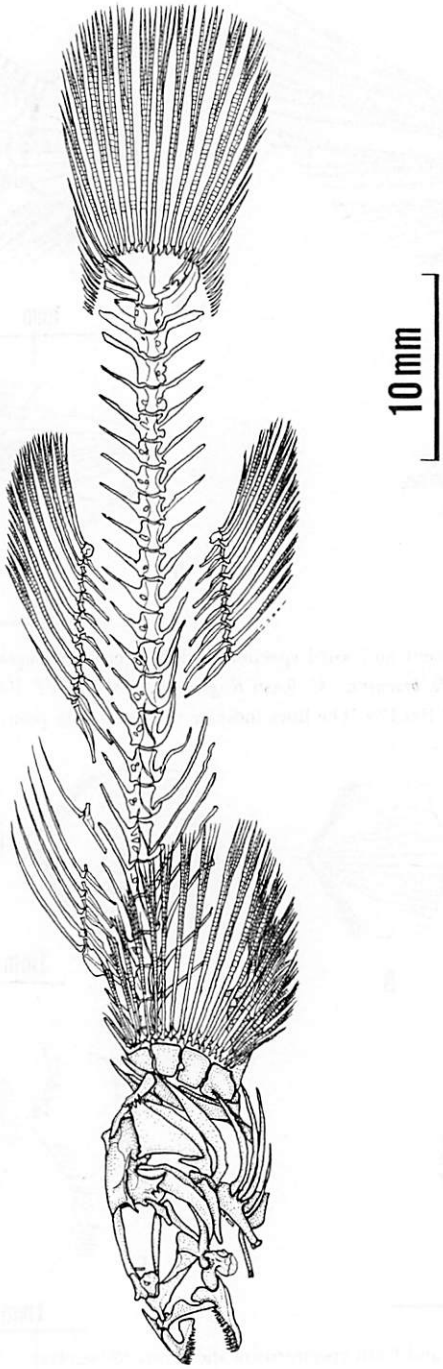

Fig. 6. Figure of the skeleton of Recent *Rhinogobius brunneus*, large dark type, KMNH VP 100,094.

observable in the fossil specimens from Kusu Basin, because the posterior part of the pelvic fin is not preserved. In Recent *R. giurinus* and *R. brunneus*, there is a difference in the position of the branching point of the innermost ray of the pelvic fin. The branching point of the innermost ray is approximately in between the base and the branching point of the next ray in *R. giurinus* (Fig. 4A). The branching point of the innermost ray is beside to the branching point of the next ray in *R. brunneus* (Fig. 4B). This character is observable in the fossil specimens (Fig. 4C and D).

In *R. giurinus*, the posterior end of the scale is slightly pointed and the spines are short on the posterior corner and gradually become longer toward the dorsal and ventral ends of the posterior margin. Grooves extend close to the posterior margin of the scale (Fig. 5B). In *R. brunneus*, the posterior margin of the scale is round in shape and the spines are almost equal in length. Grooves extend two thirds of the way from the anterior end (Fig. 5C). In *R. giurinus*, scales similar to those of *R. brunneus* occur in a ratio of approximately one to twenty (Fig. 5A). In *R. brunneus*, scales similar to those of *R. giurinus* occur in a ratio of one to five or six (Fig. 5D). Two types of scales are recognized in the fossil specimens (Fig. 5E and F). All of the observable scales in a fossil specimen are examined for identification of species. Two species, *R. giurinus* and *R. brunneus*, are recognized among fossil specimens on the bases of the characters of the pelvic fin and scales (Figs. 4C and D, 5E and F).

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Pleistocene Gobiid Fishes of the Genus *Rhinogobius* from Kusu Basin, Oita Prefecture, Japan.

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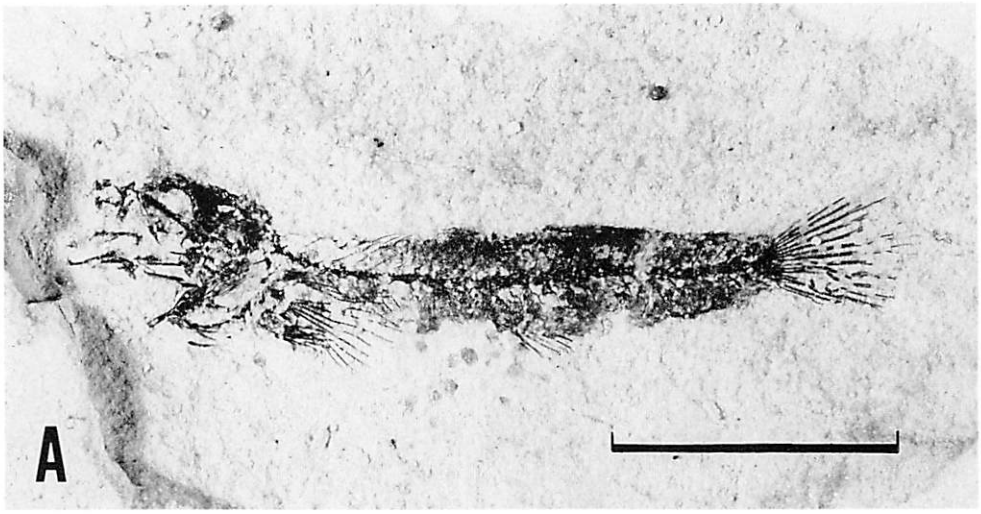
Plates 1–2

Explanation of Plate 1.

Photographs of the fossils of *Rhinogobius giurinus* (RUTTER) from Kusu Basin, Oita Prefecture.

A. lateral view of the left side (KMNH VP 100,118).
B. ventral view (KMNH VP 100,119).
C. dorsal view (KMNH VP 100,117).
Scales indicate 10 mm.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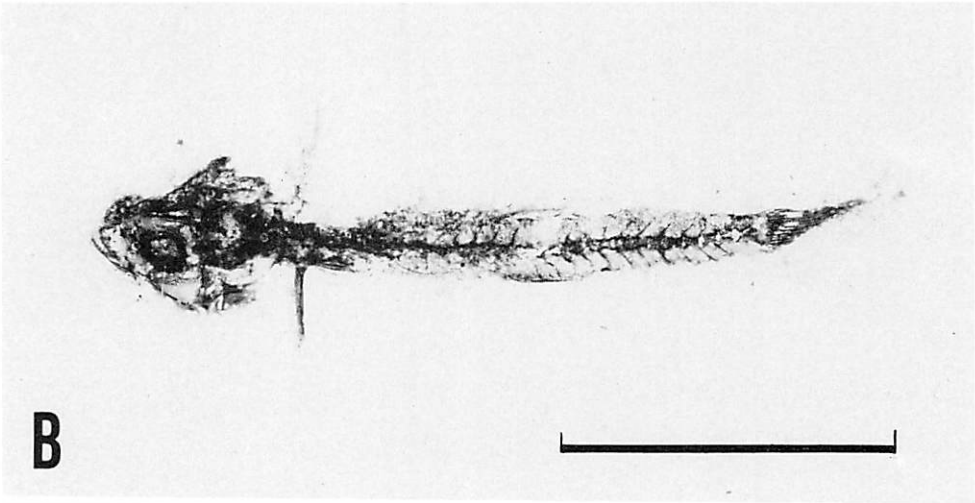
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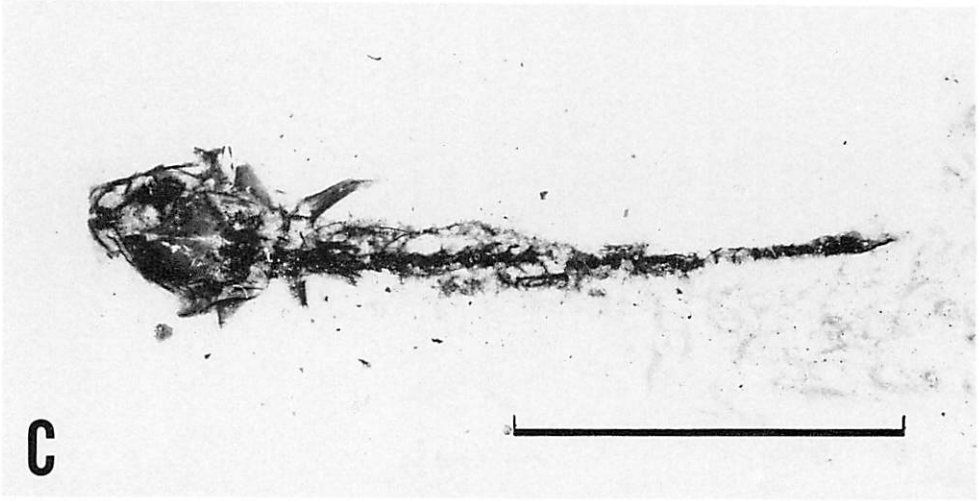
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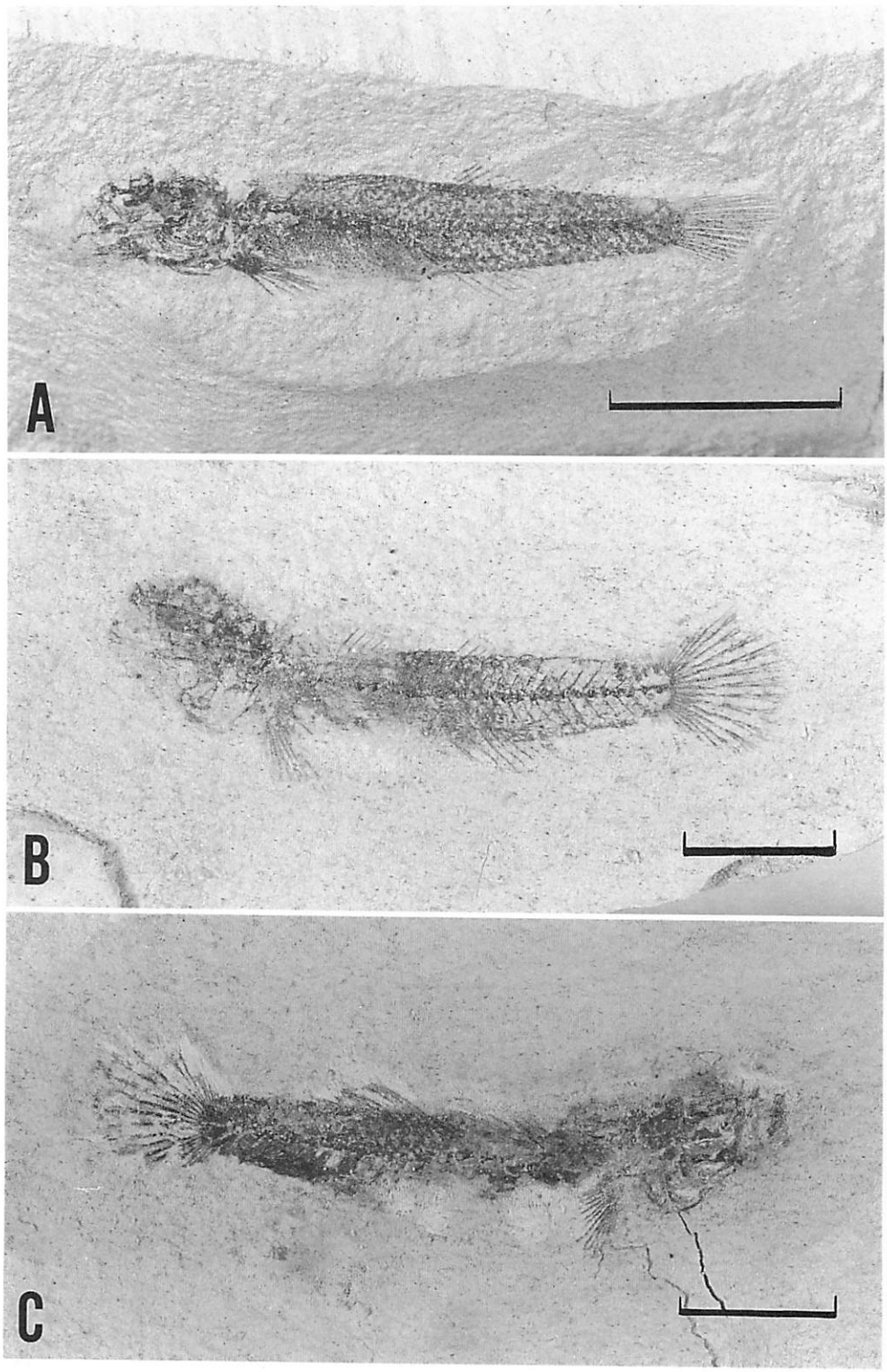
Plate 2

Explanation of Plate 2.

Photographs of the fossils of *Rhinogobius brunneus* (TEMMINCK et SCHLEGEL) from Kusu Basin, Oita Prefecture.

A. lateral view of the left side (KMNH VP 100,125).
B. lateral view of the left side (KMNH VP 100,129).
C. lateral view of the right side (KMNH VP 100,132).

Scales indicate 10 mm.


A
B
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